Model-data comparisons / Coupled modelling

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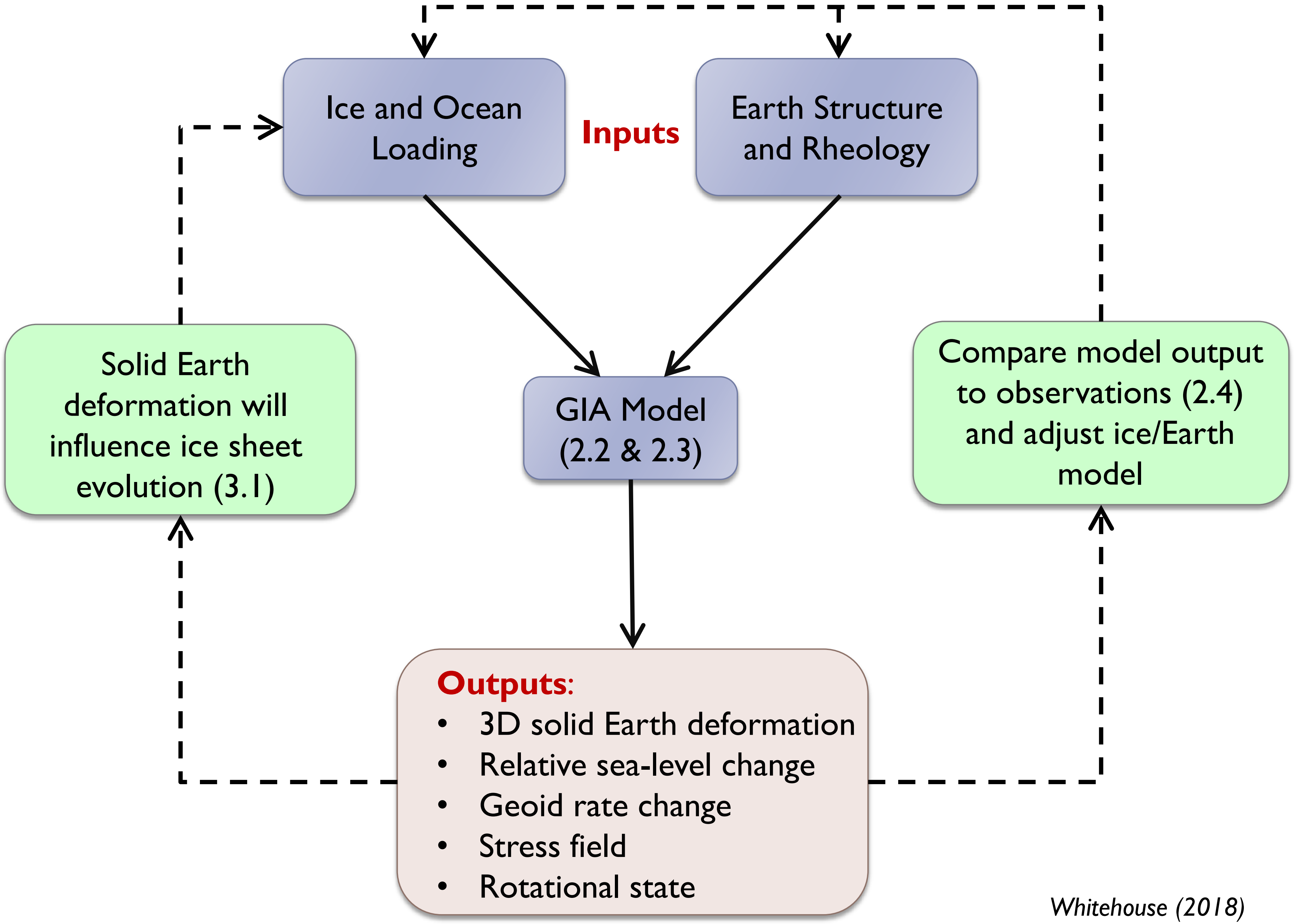
Part 1: model-data comparisons

- download the TurningPoint app

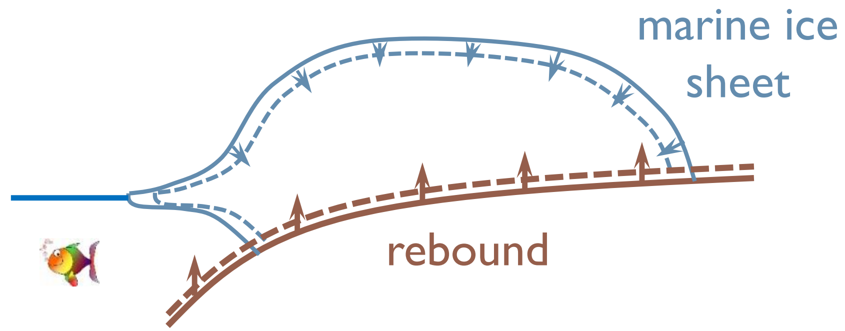
- recall earlier material on
  - GIA model inputs
  - tuning a GIA model
  - data constraints on model inputs

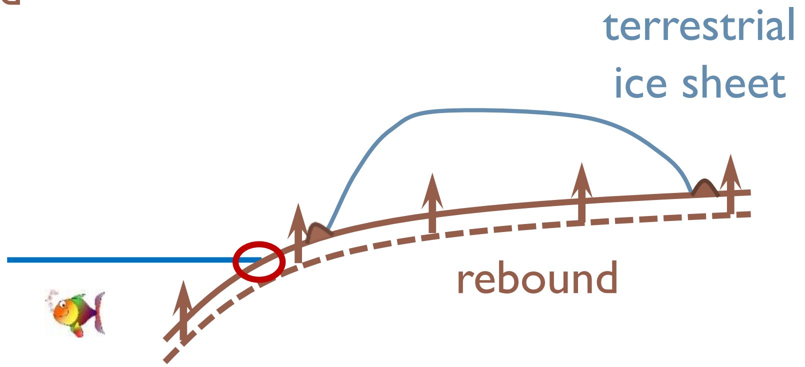
GIA Model

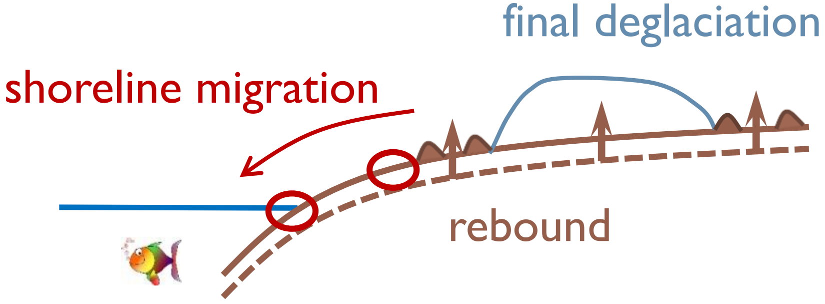


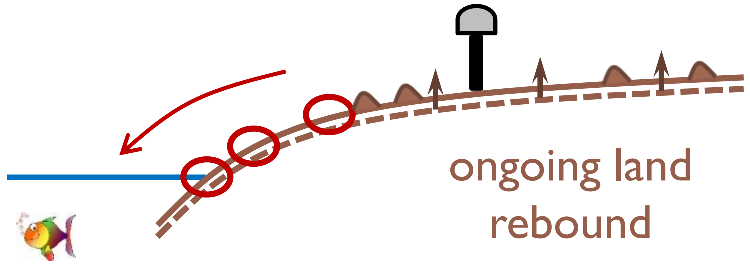
Whitehouse (2018)

Determining 'GIA unknowns'

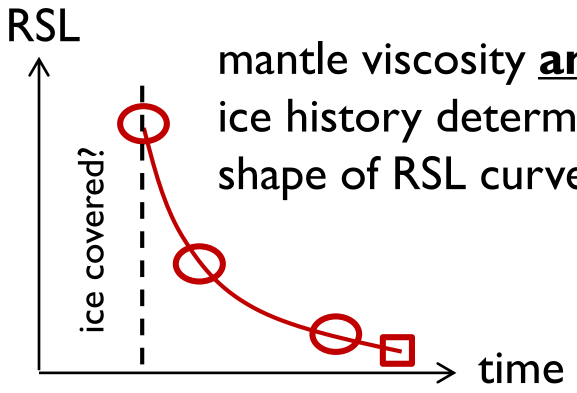




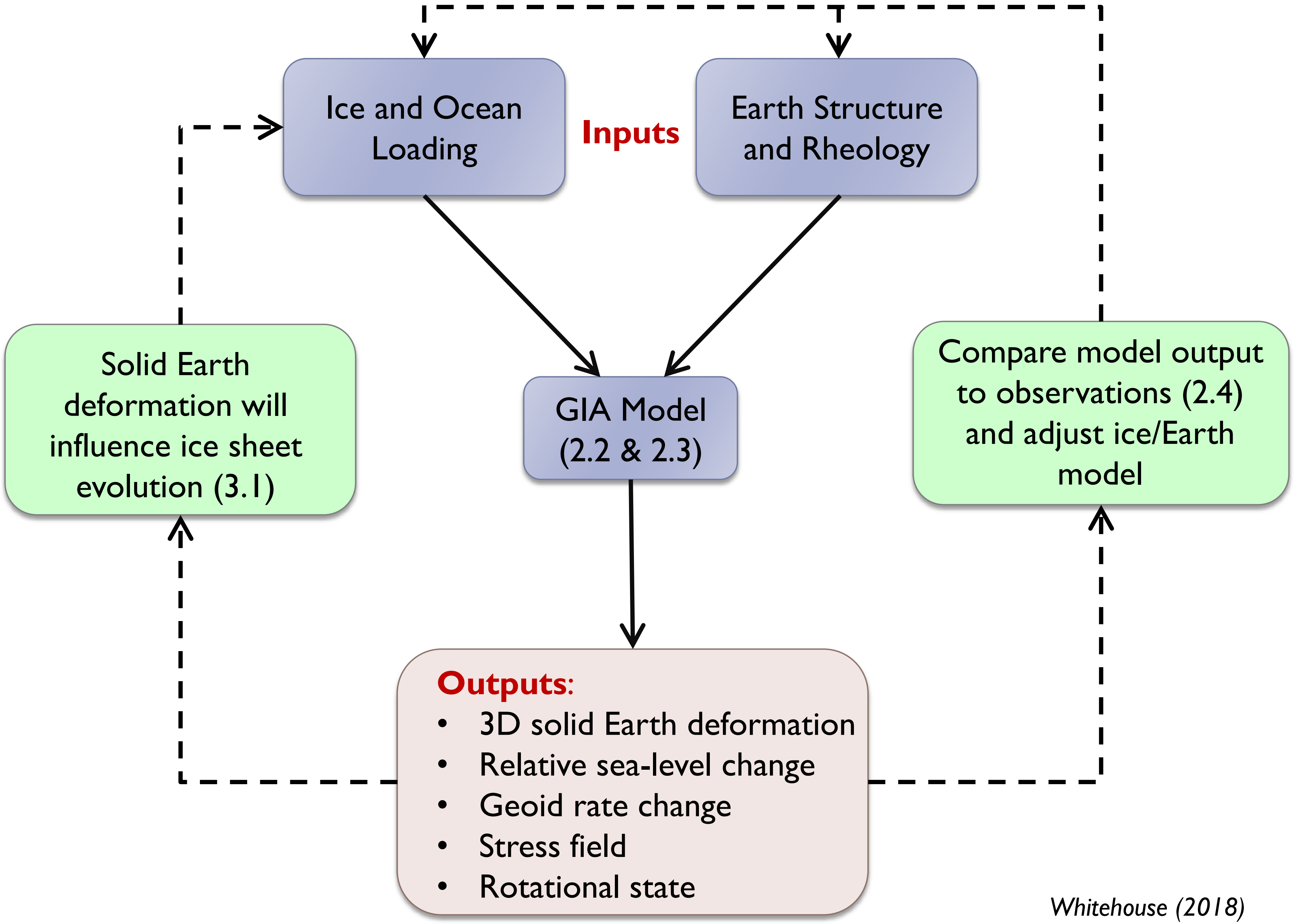




mantle viscosity **and** ice history determine shape of RSL curve



GIA Model



Whitehouse (2018)

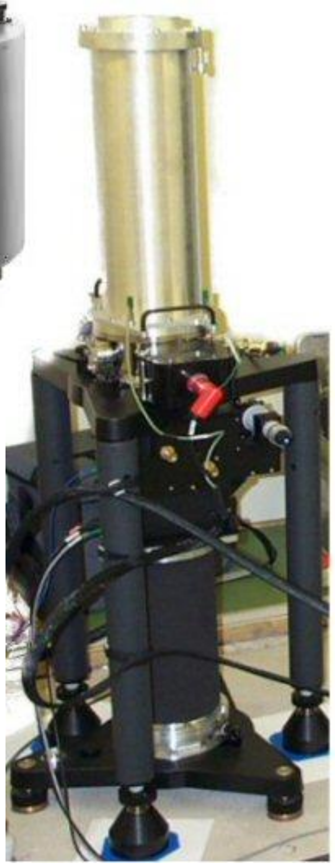
Data…

sea-level change

ice sheet change

earth deformation

geophysics…

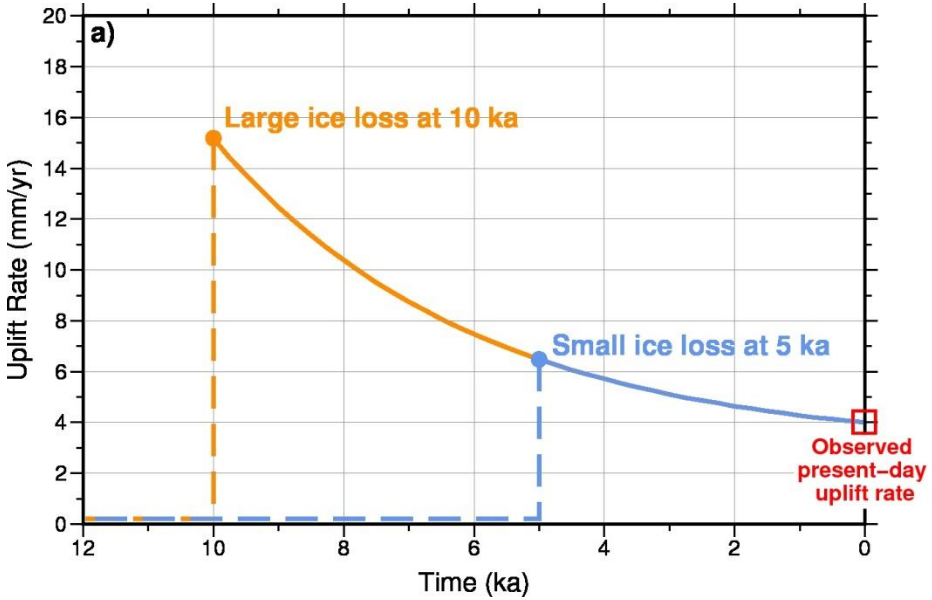


Non-uniqueness in GIA

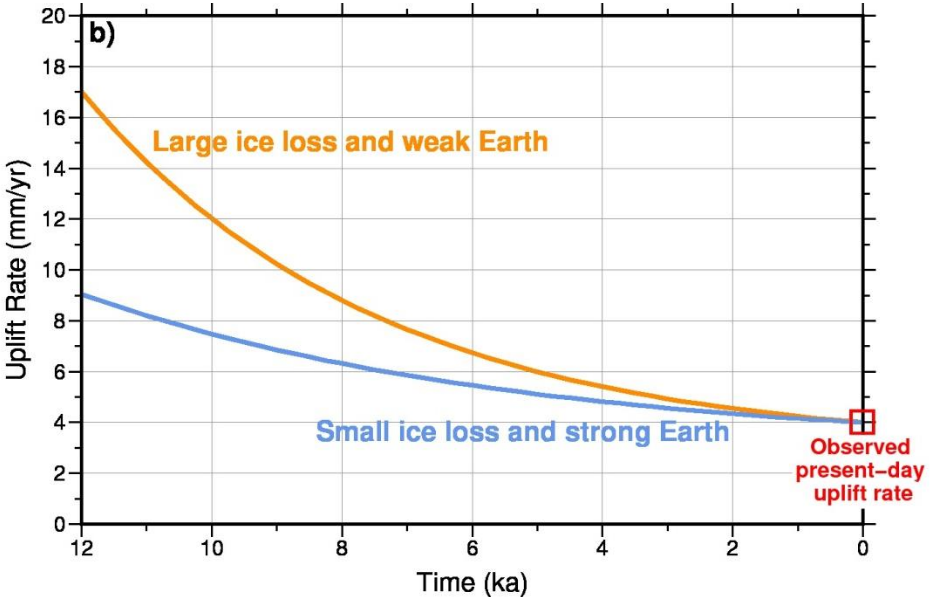
Trade-off between:

- <u>timing</u> of ice loss
- <u>magnitude</u> of ice loss
- <u>relaxation time</u> of the solid Earth (rheology)

<u>Known</u>: mantle relaxation time
<u>Unknown</u>: timing & magnitude of ice loss



<u>Known</u>: timing of ice loss
<u>Unknown</u>: magnitude of ice loss & mantle relaxation time

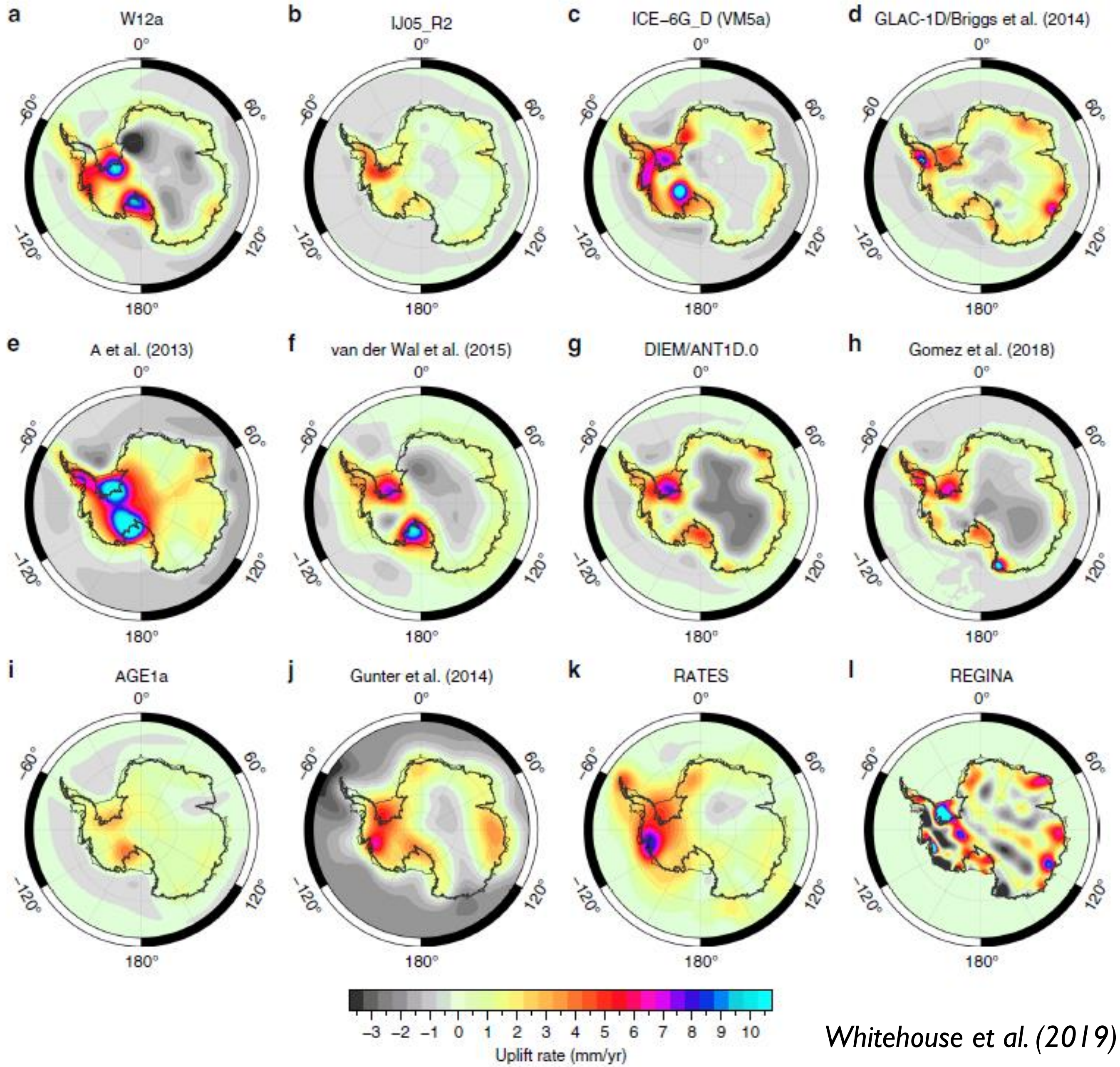


Whitehouse (2018)

Determining the 'GIA signal'

- Predictions of uplift rates across Antarctica due to GIA

- (a-d) 1D GIA model
- (e-f) 3D GIA model (Friday)
- (g-h) coupled GIA model (part 2)
- (i-l) data inversion



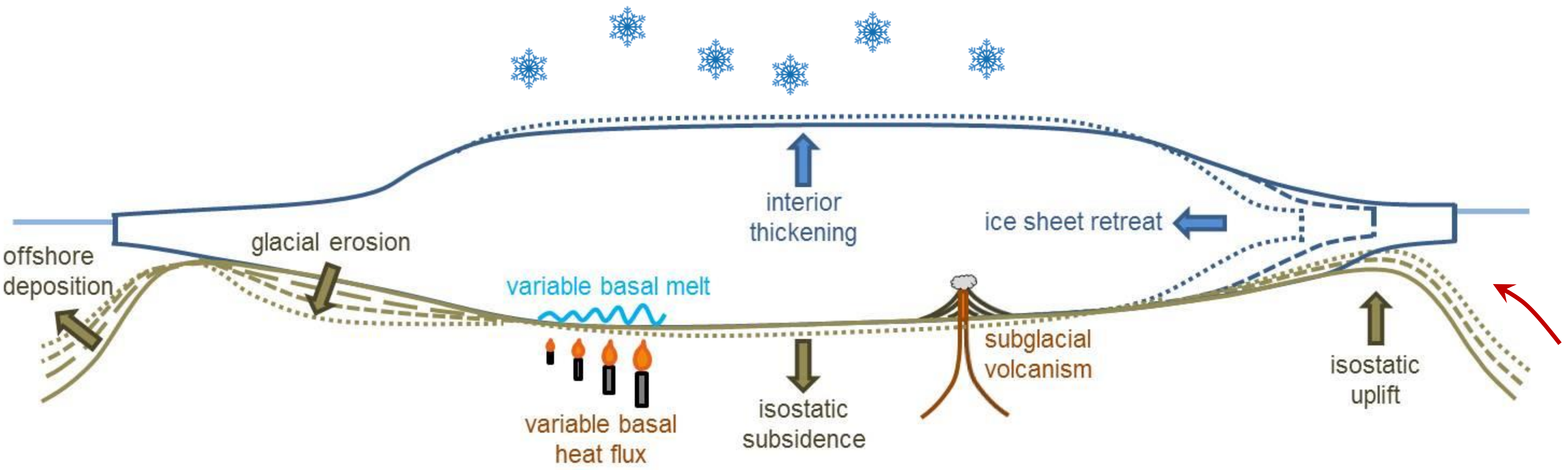
Whitehouse et al. (2019)

Part 2: coupled modelling

Hello, I'd like to use you as an input

Well, actually I need some input from you first

Ice sheet – solid Earth interactions



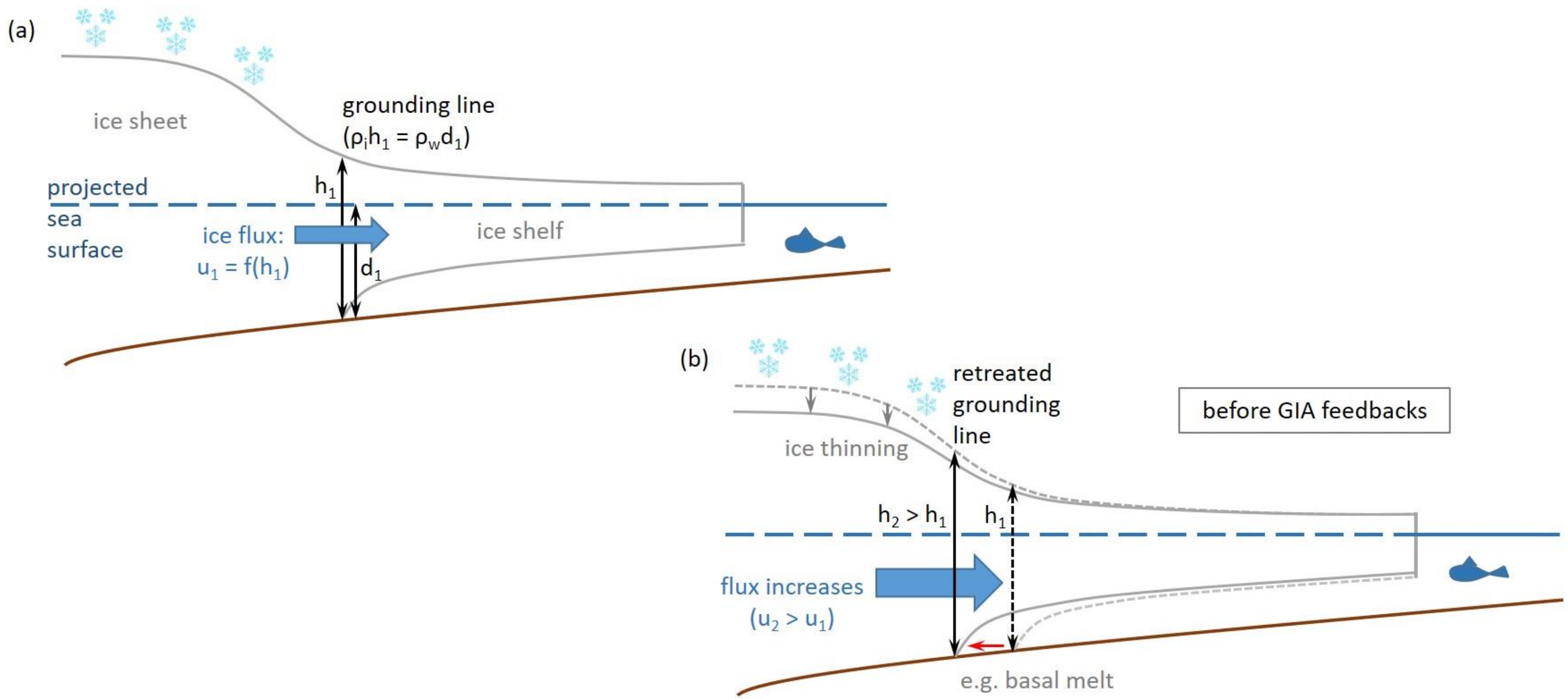
<u>Ice sheet change</u>:

- deforms the solid Earth
- alters sea level

} how do these affect ice sheet change?

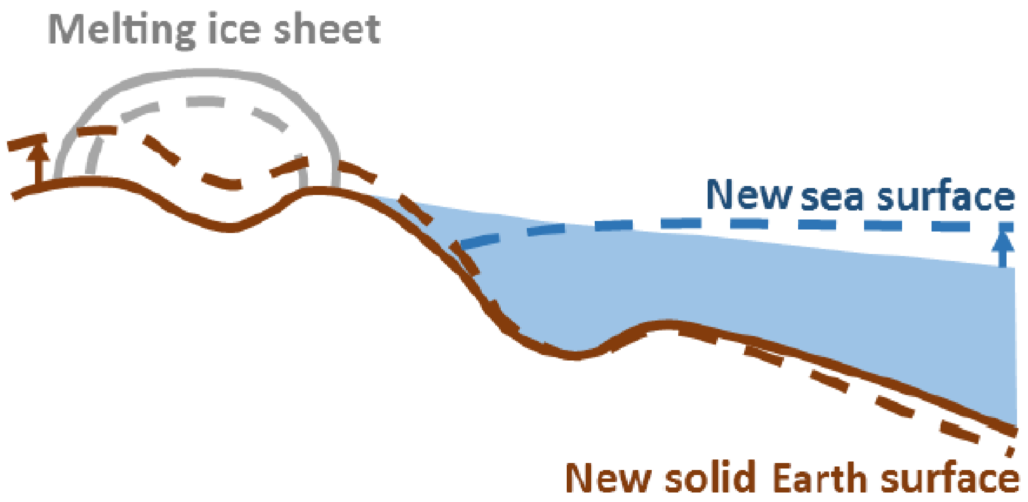
Whitehouse et al. (2019)

Ice sheet dynamics



Whitehouse et al. (2019)

Spatially variable sea-level change

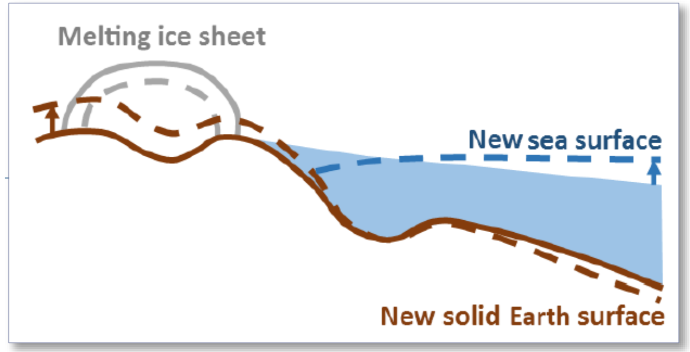


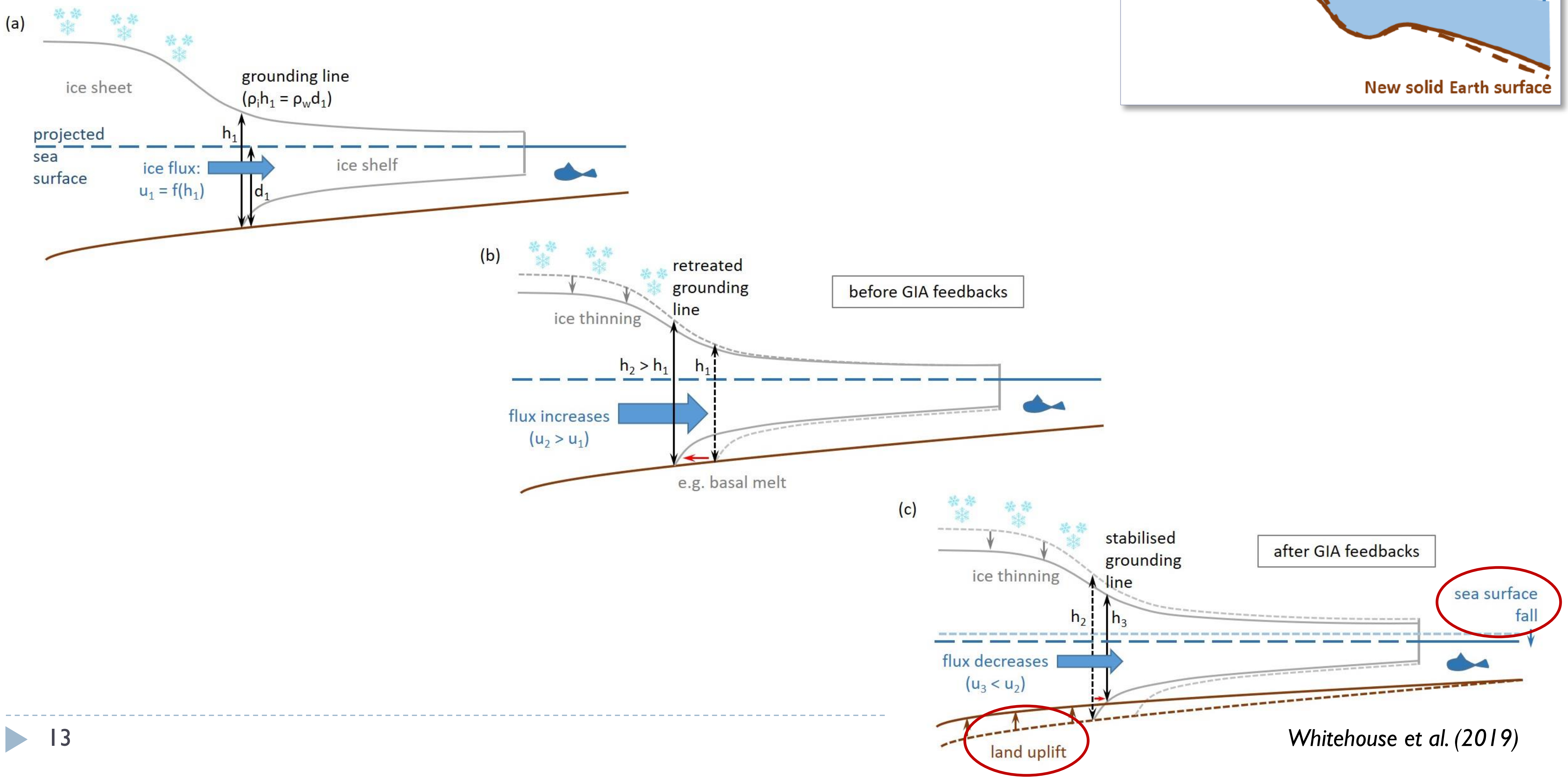
- loss of ice mass
- land uplift
- perturbation to shape of geoid
- elastic and viscous processes

- Water depth adjacent to the grounding line decreases as the ice sheet shrinks
- The rate, magnitude, and spatial pattern depend on Earth rheology

Whitehouse (2018), after Conrad (2013)

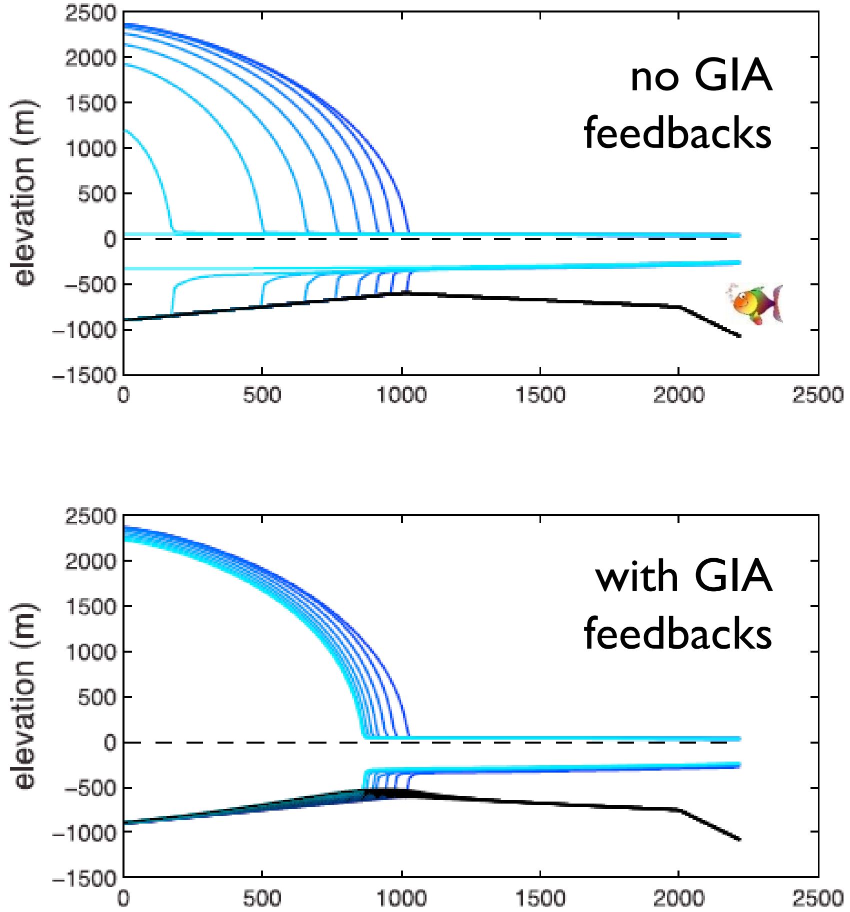
GIA – ice dynamic feedbacks





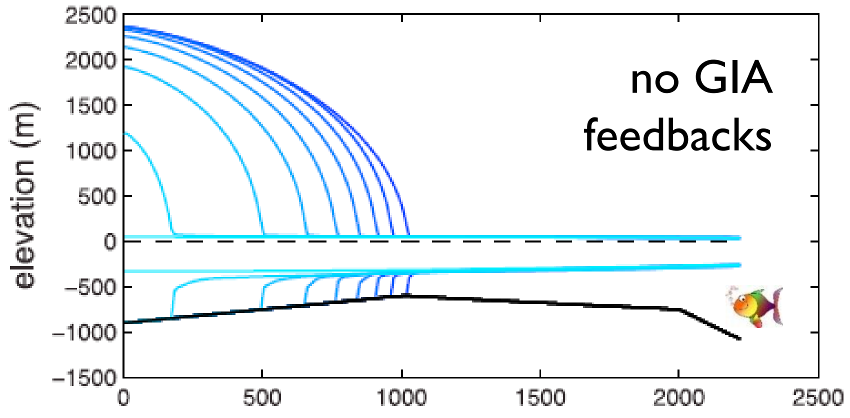
Whitehouse et al. (2019)

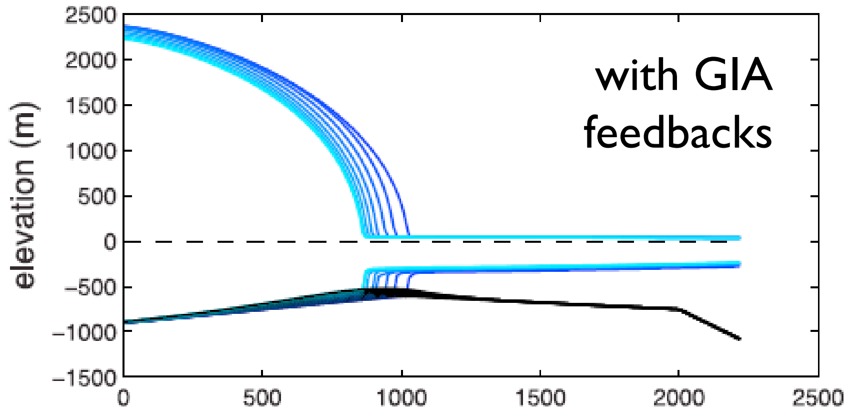
Does including GIA effects alter ice sheet dynamics?



Gomez et al. (2012)

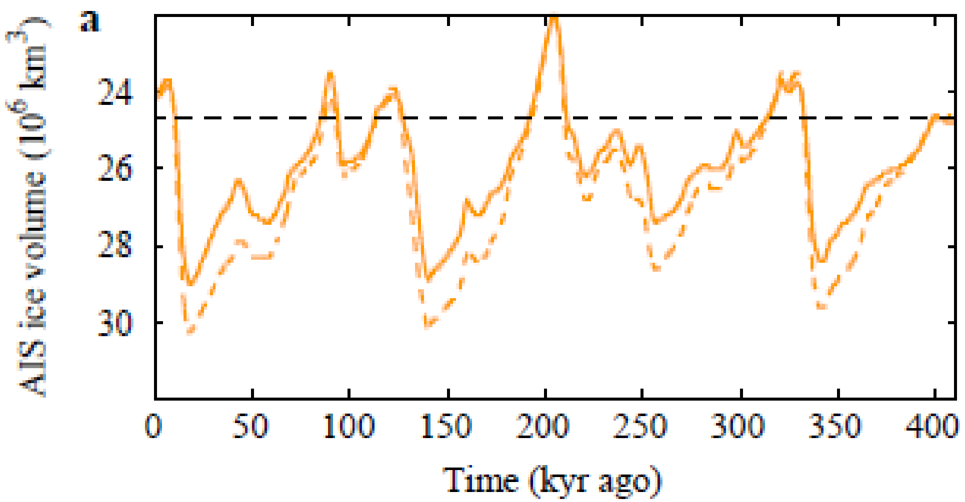
Does including GIA effects alter ice sheet dynamics?





Gomez et al. (2012)

Solid line = coupled model
Dashed line = forced with eustatic SL



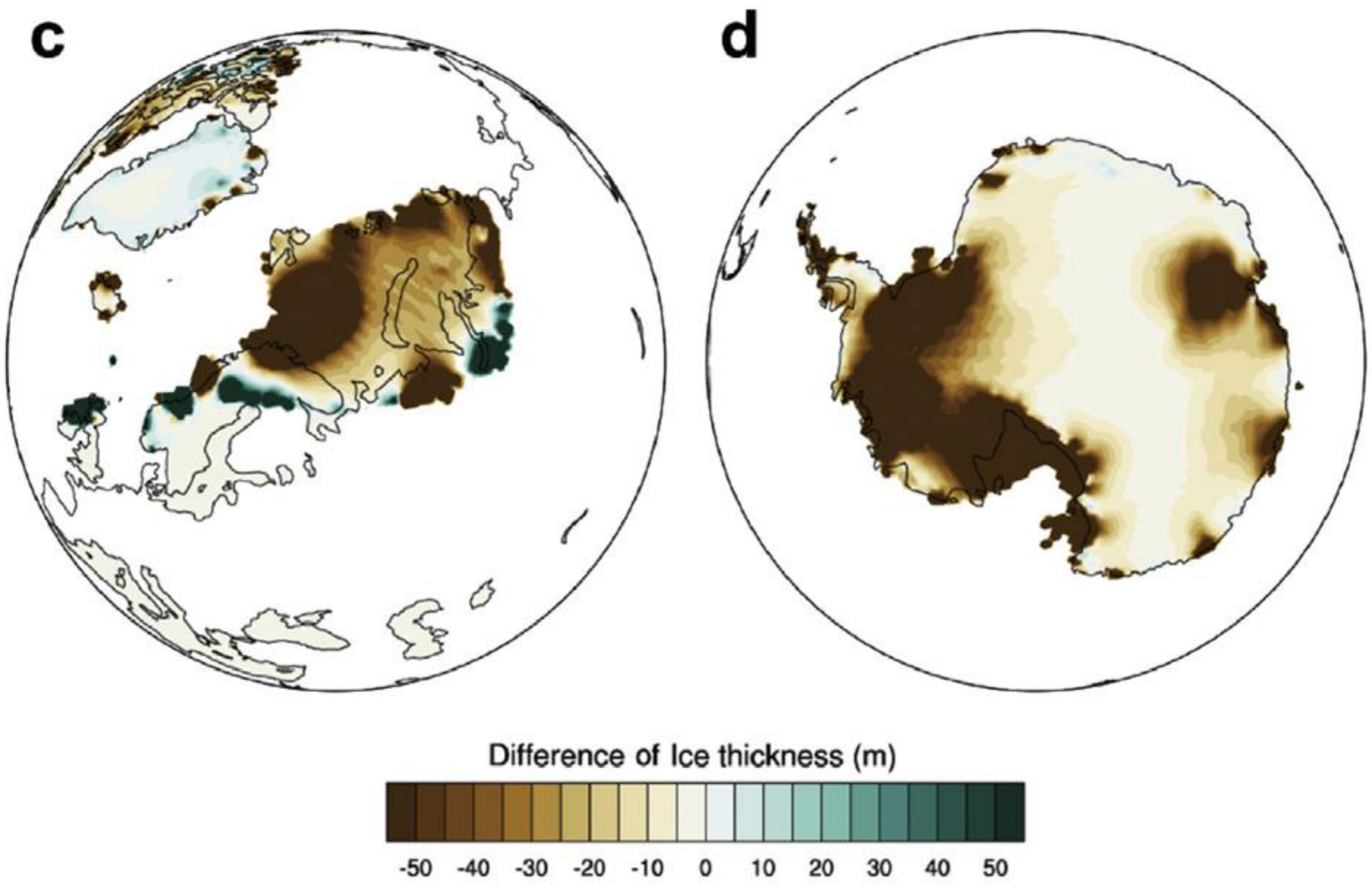
de Boer et al. (2014)

- Damps the rate of grounding line retreat (or advance)
- Reduces the magnitude of glacial-interglacial variability

Does including GIA effects alter ice sheet dynamics?

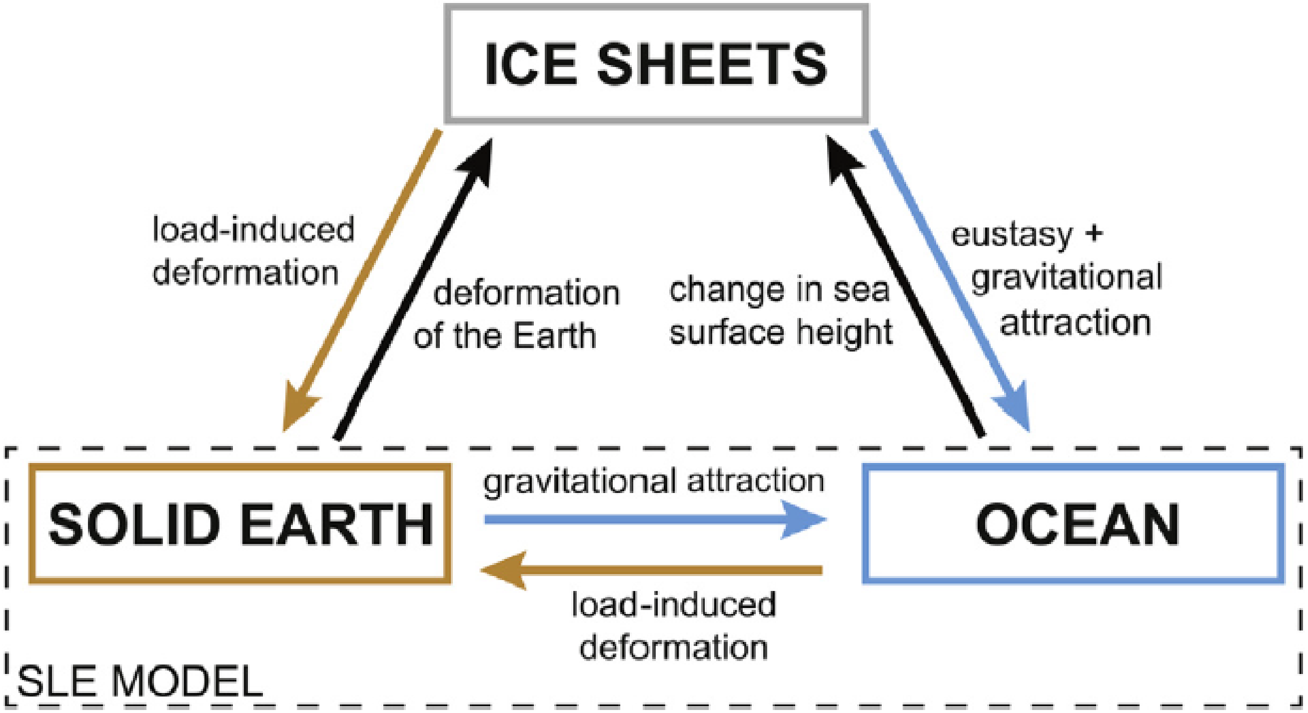
- Difference between ice thickness at the glacial maximum modelled by a coupled model (spatially variable SL forcing) and an uncoupled model (uniform SL forcing)

- Same earth model



de Boer et al. (2014; 2017)

GIA – ice sheet coupled modelling

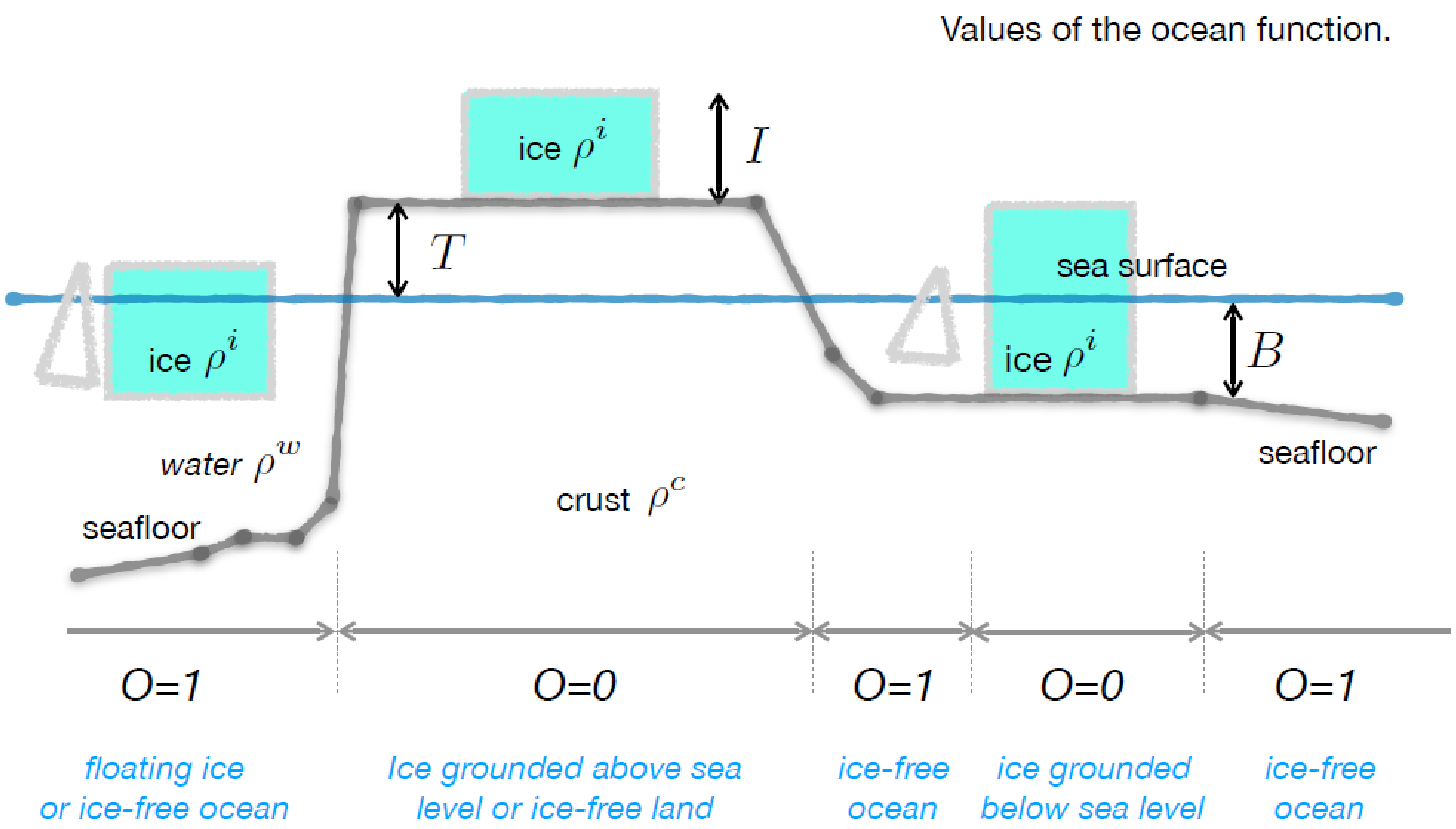


Complications

- the SLE must be solved iteratively
- viscosity and time
- GIA/ice models use different spatial resolution
- absolute water depth is important
  - "For example, a GIA model could be used to calculate the change in RSL between 20 and 10 kyr ago, but without knowing how RSL changed between 10 kyr ago and the present is it not possible to relate the changes between 20 and 10 kyr ago to absolute water depths. Crucially, absolute water depth is the variable that feeds into the ice-sheet model."
  - Also important for ocean function

de Boer et al. (2017)

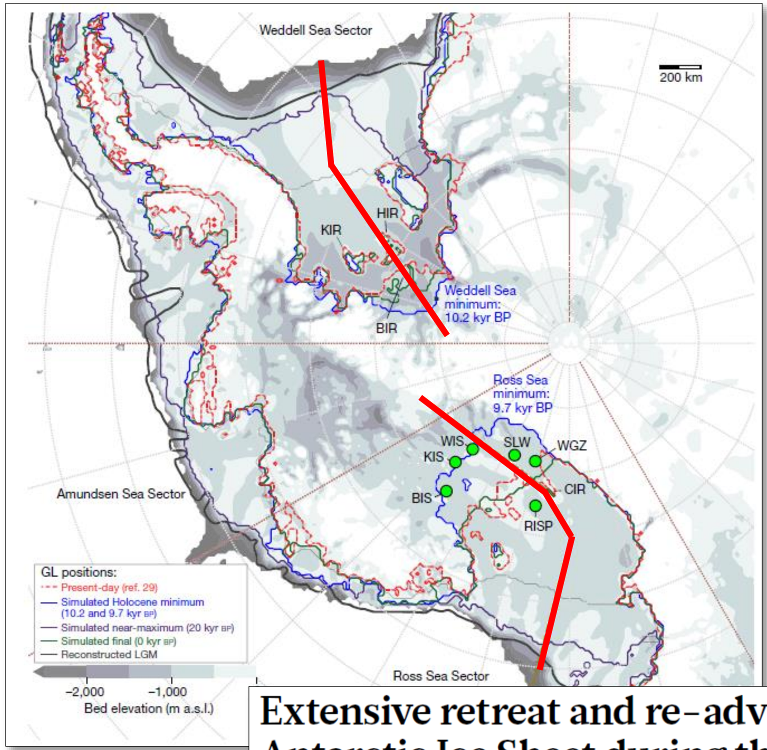
Ocean function



From Giorgio Spada's lecture

But are GIA – ice dynamic feedbacks important in reality?

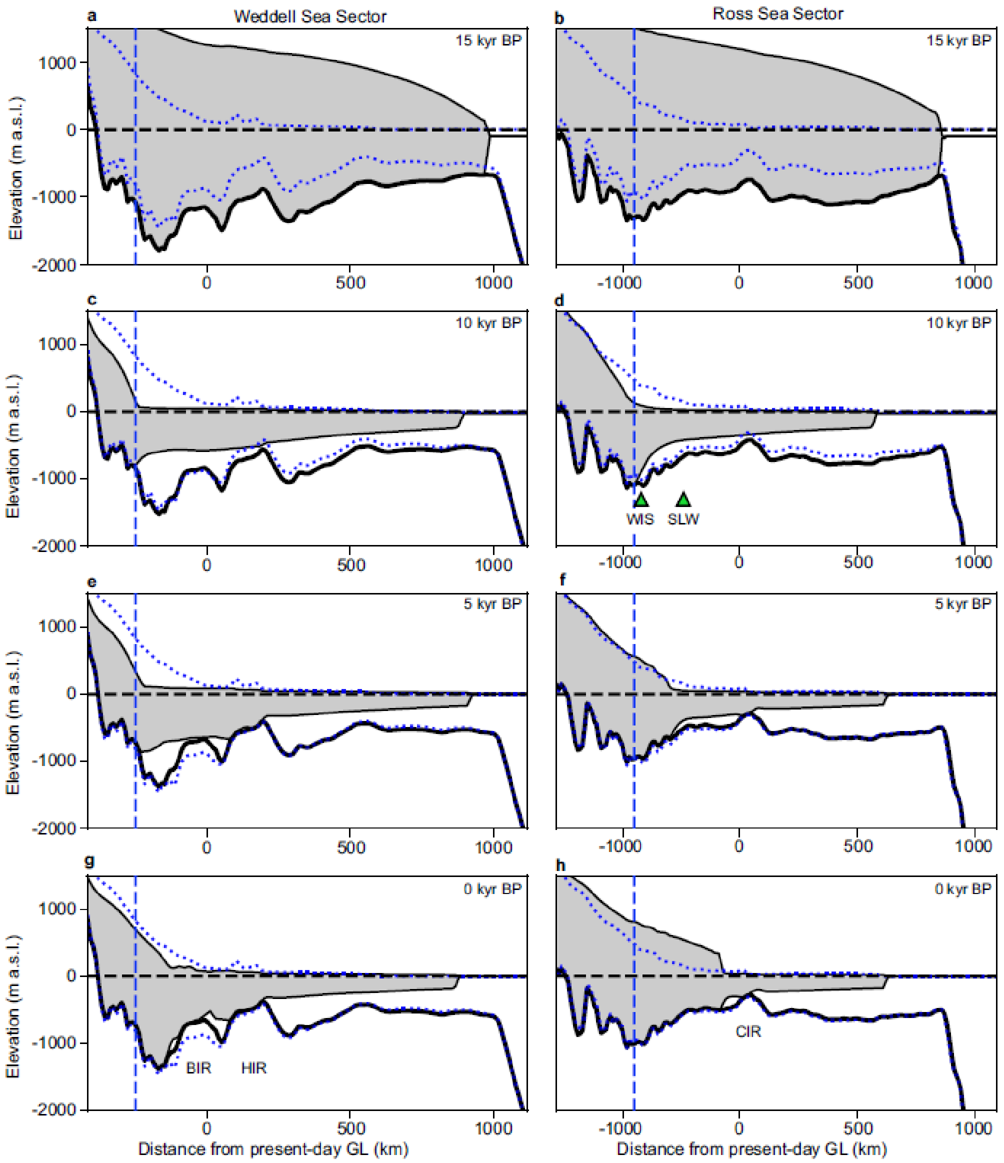
Stabilizing effect of GIA in the past…



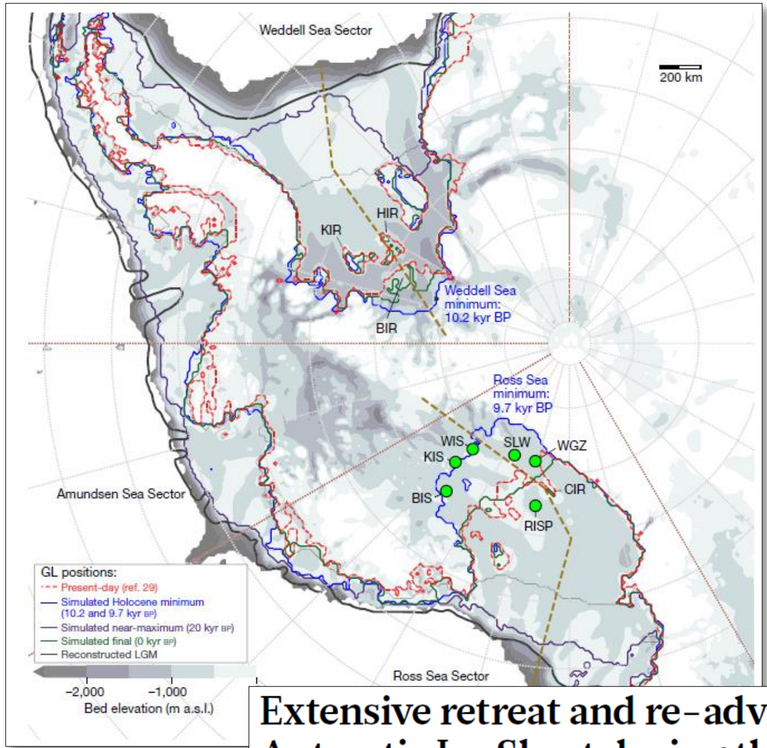
Extensive retreat and re-advance of the West Antarctic Ice Sheet during the Holocene

J. Kingslake[1,6*], R. P. Scherer[2,6], T. Albrecht[3,6], J. Coenen[2], R. D. Powell[2], R. Reese[3], N. D. Stansell[2], S. Tulaczyk[4], M. G. Wearing[1] & P. L. Whitehouse[5]

- Solid Earth rebound is hypothesized to have triggered ice sheet re-advance during the Holocene



Stabilizing effect of GIA in the past and the future?



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- Solid Earth rebound is hypothesized to have triggered ice sheet re-advance during the Holocene

- Model predictions demonstrate that GIA can play a role in controlling future ice sheet dynamics <u>if mantle viscosity is low enough</u>

ARTICLE

Received 2 Jun 2015 | Accepted 6 Oct 2015 | Published 10 Nov 2015 DOI: 10.1038/ncomms9798 OPEN

Sea-level feedback lowers projections of future Antarctic Ice-Sheet mass loss

Natalya Gomez[1,2], David Pollard[3] & David Holland[1]

Contents lists available at ScienceDirect
ELSEVIER
Earth and Planetary Science Letters
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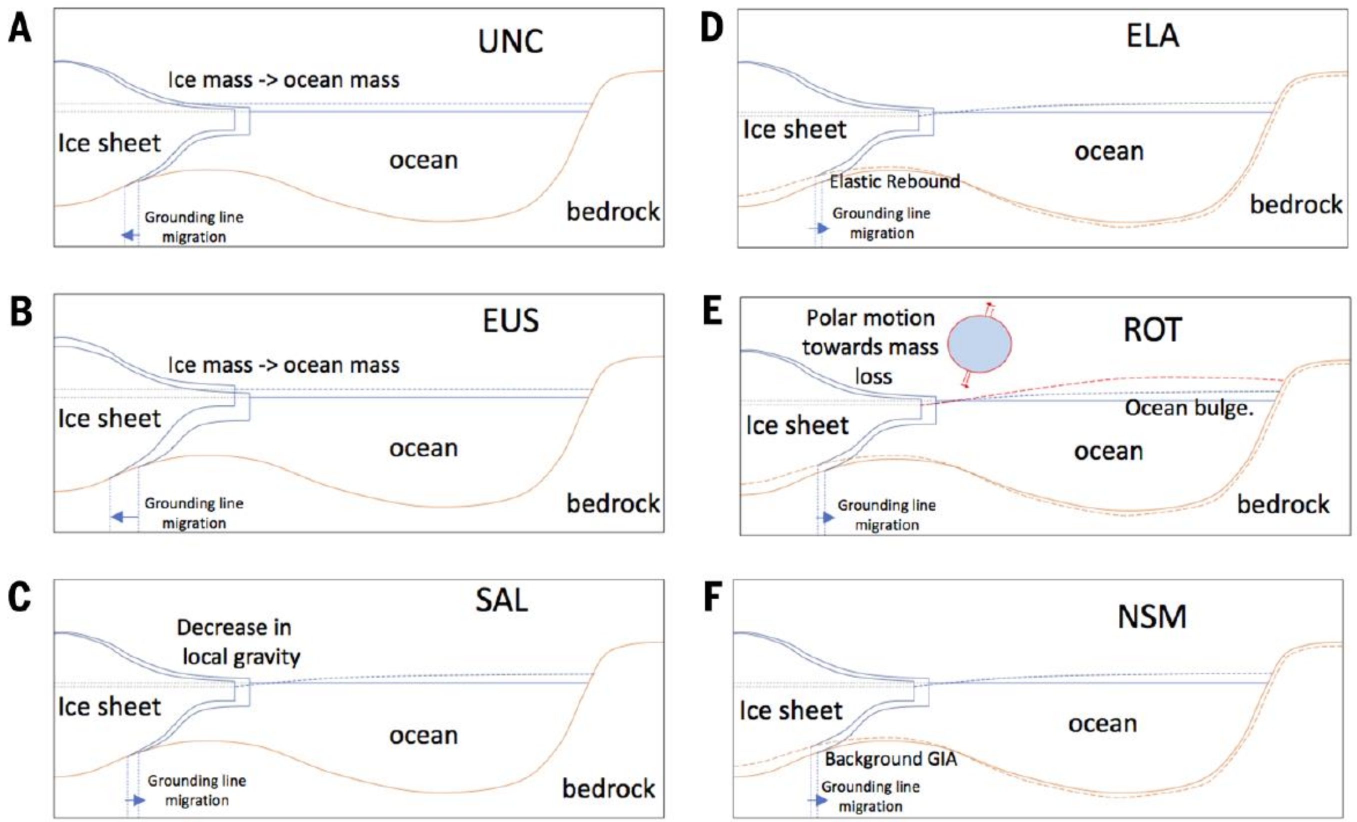
Potential of the solid-Earth response for limiting long-term West Antarctic Ice Sheet retreat in a warming climate

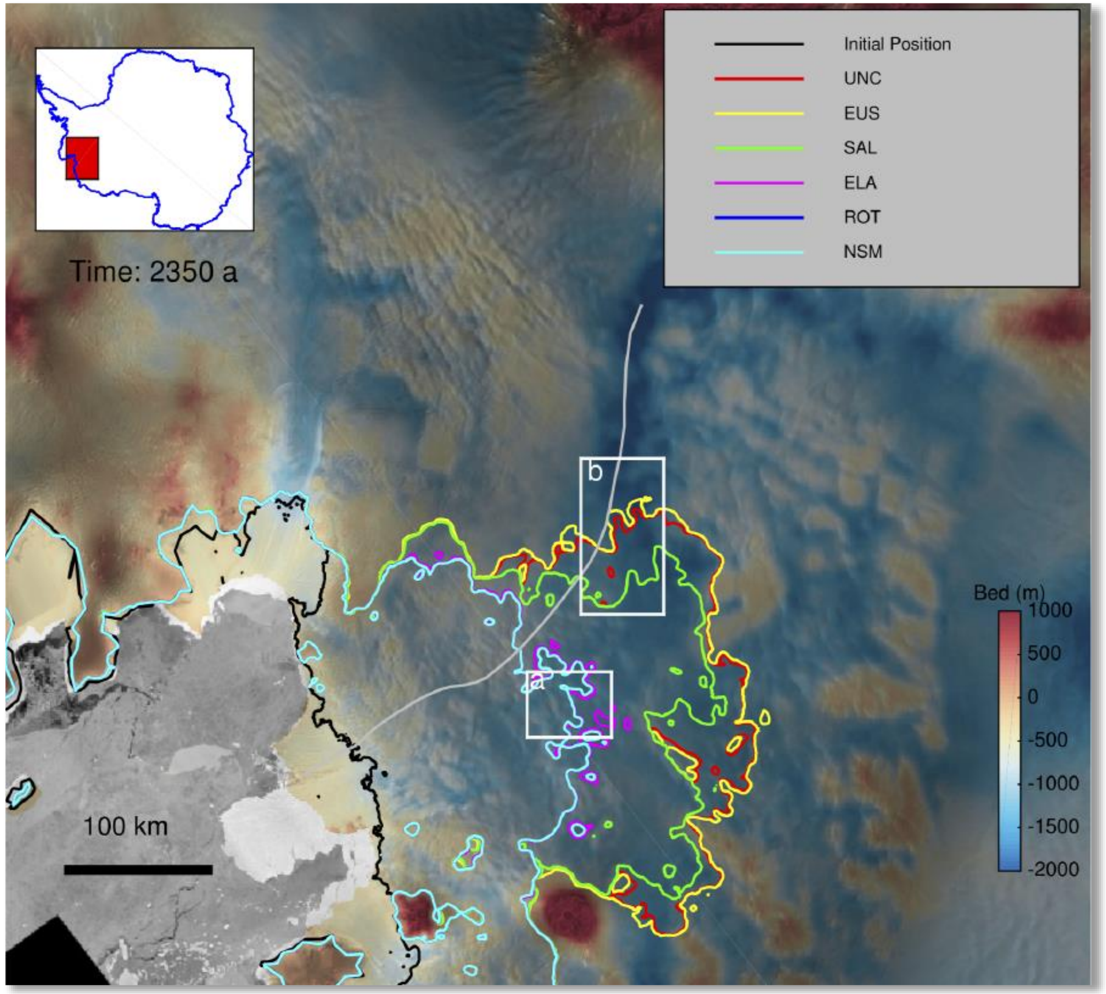
CrossMark

Hannes Konrad[a,b,c,*,1], Ingo Sasgen[a,d], David Pollard[e], Volker Klemann[a]

Slowdown in Antarctic mass loss from solid Earth and sea-level feedbacks

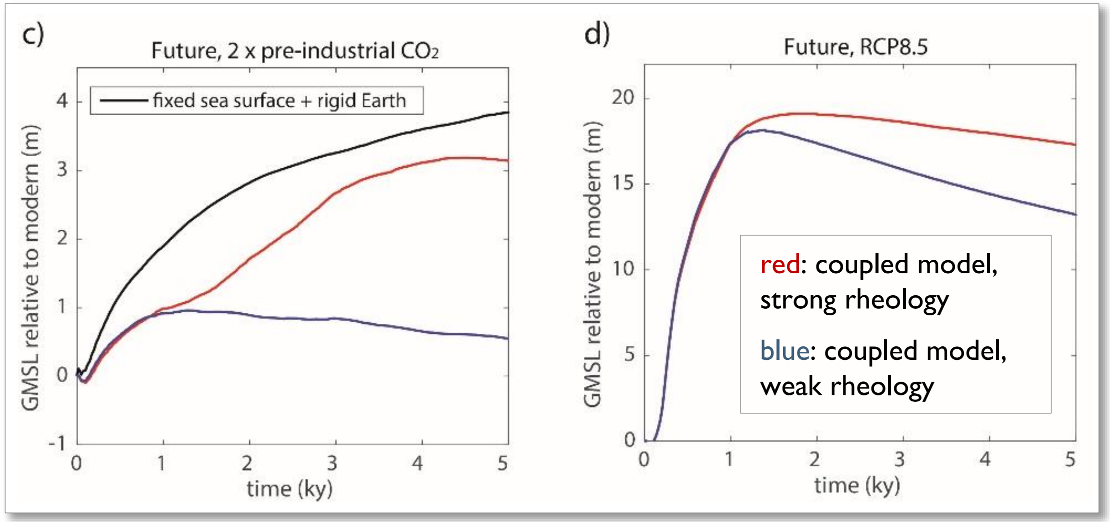
E. Larour*, H. Seroussi, S. Adhikari, E. Ivins, L. Caron, M. Morlighem, N. Schlegel





What is the magnitude of the stabilizing effect of GIA?

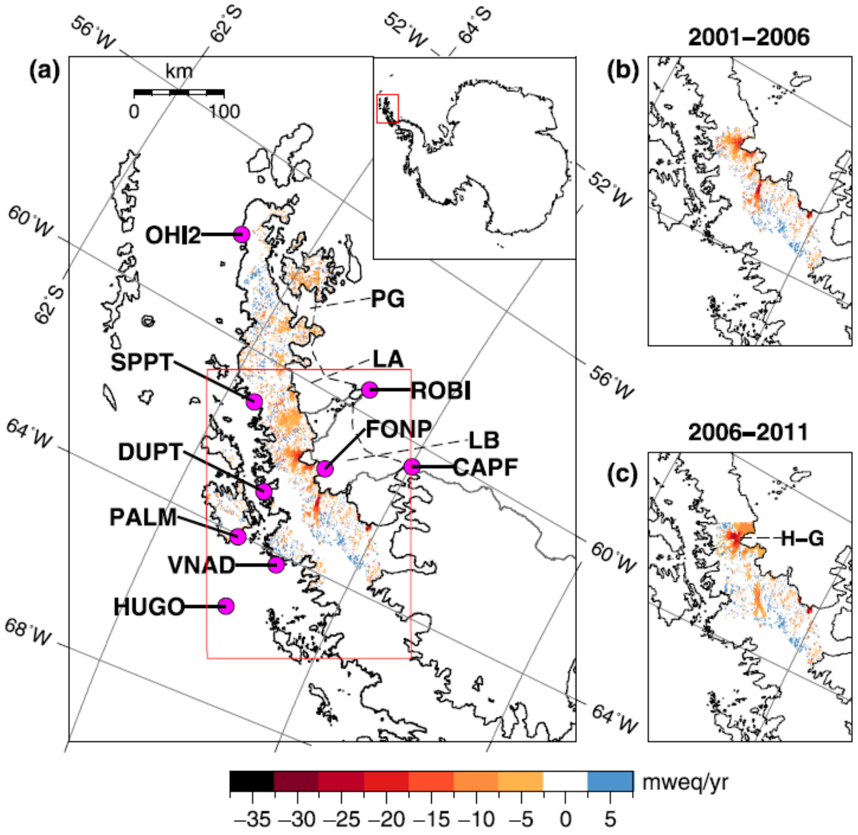
- Hypothesize that GIA can play a role in damping the rate of future ice loss if mantle viscosity is low enough
- Need to determine strength of feedbacks

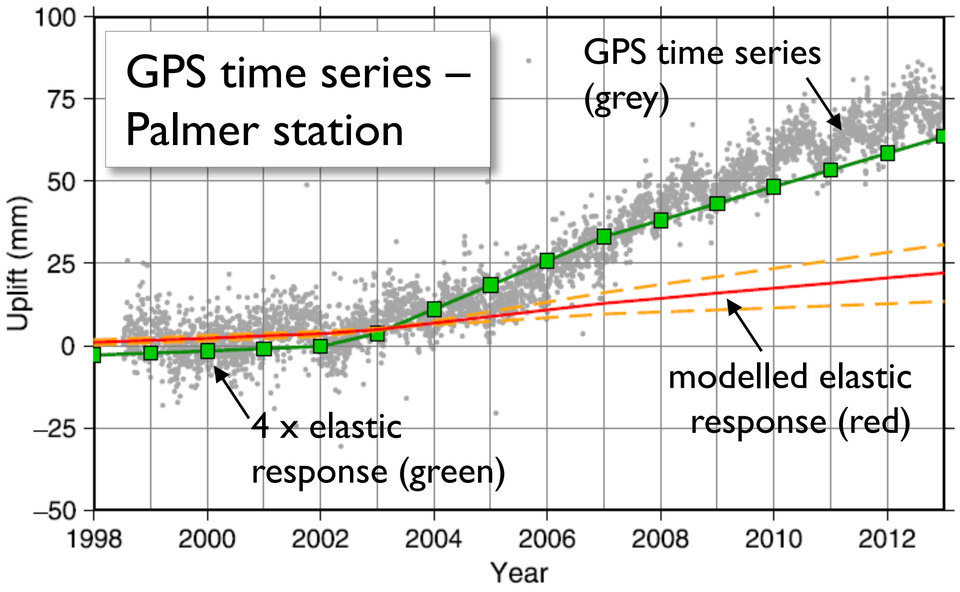


Whitehouse et al. (2019) [adapted from Pollard et al. (2017)]

What is the mantle viscosity beneath Antarctica?

- Natural experiment: Larsen B ice shelf break-up
- Known ice history (left - from altimetry) and rebound (right - from GPS)



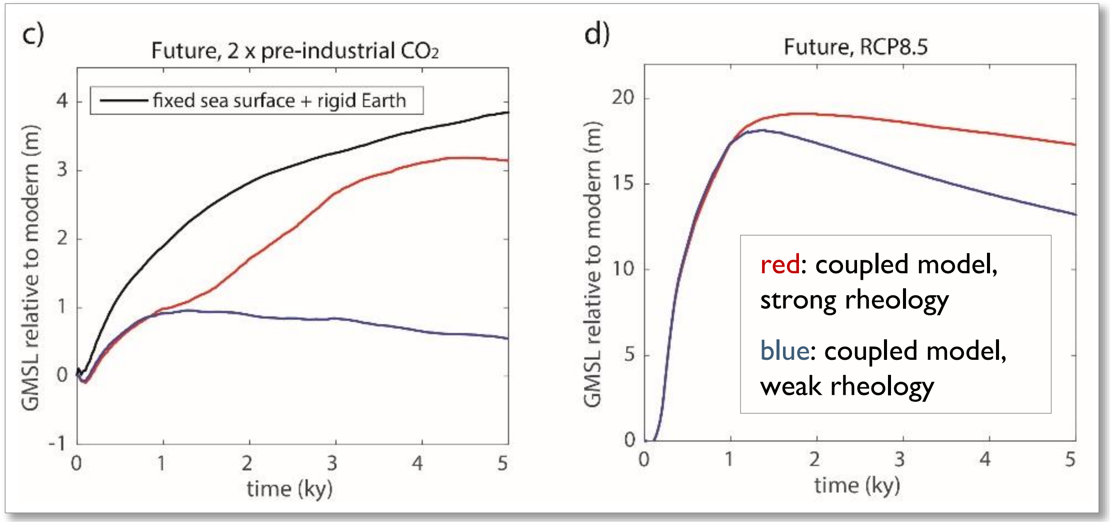


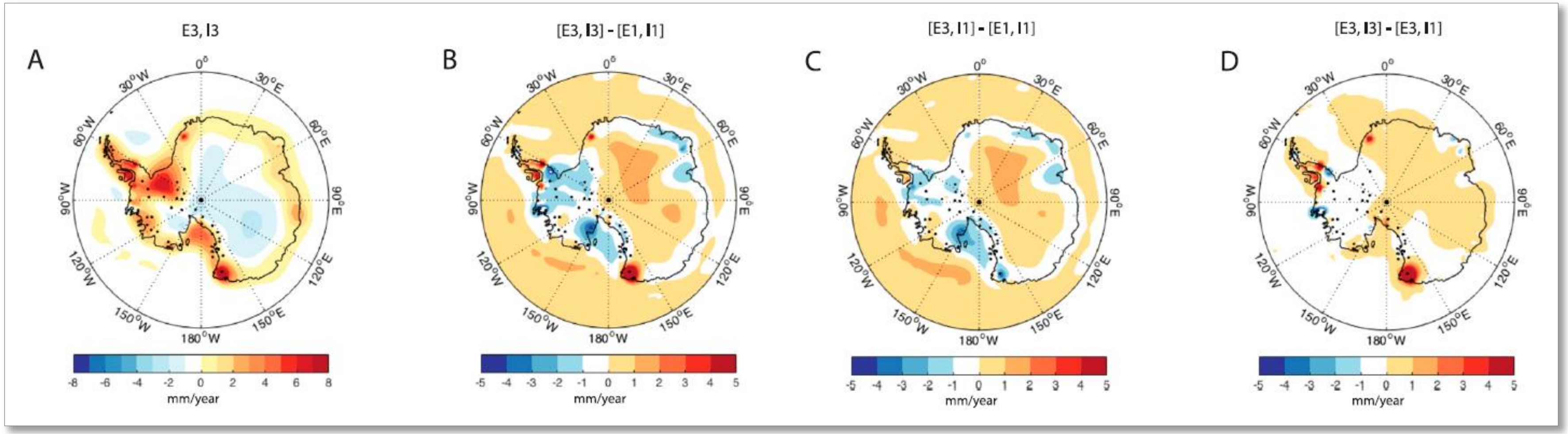
Solid Earth response implies upper mantle viscosity of 6 x 10^{17} – 2 x 10^{18} Pa s

Nield et al. (2014)

What is the magnitude of the stabilizing effect of GIA?

- Hypothesize that GIA can play a role in damping the rate of future ice loss if mantle viscosity is low enough
- Need to determine strength of feedbacks
- What next? – a coupled GIA-ice dynamics model that accounts for 3D variations in Earth rheology





top: Whitehouse et al. (2019) [adapted from Pollard et al. (2017)], bottom: Gomez et al. (2018)

Summary

- Part 1
  - different types of data constrain different components of a GIA model
  - many ‘GIA observations’ are sensitive to both ice history and Earth rheology

- Part 2
  - water depth plays an important role in controlling ice sheet dynamics
  - water depth change is spatially variable over time (due to GIA)
  - feedbacks between ice sheet dynamics and GIA processes can be explored using a coupled model

References (part 1)

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